MCP At the table !

Build your local server to tame the fridge chaos

Bearded tech guy with questionable facial hair

Dad

- Foster growth
- Teach & learn everyday
- Nonviolent communication

Developer

- SOLID
- Software craftsmanship
- Polyglot

Architect

- Use industry recognized norms & practices
- Make it easy to do right, hard to do wrong

Pragmatic & introspective

- Learned a lot to "keep the good" and "avoid the bad"... but mostly from "avoid the bad
- Understanding comes from experience
- Understanding evolves over time

Agilist

- Love small incréments
- Deliver fast and often
- Focus on business value
- Estimates are a lie
- Technical-to-business translator

Passionate

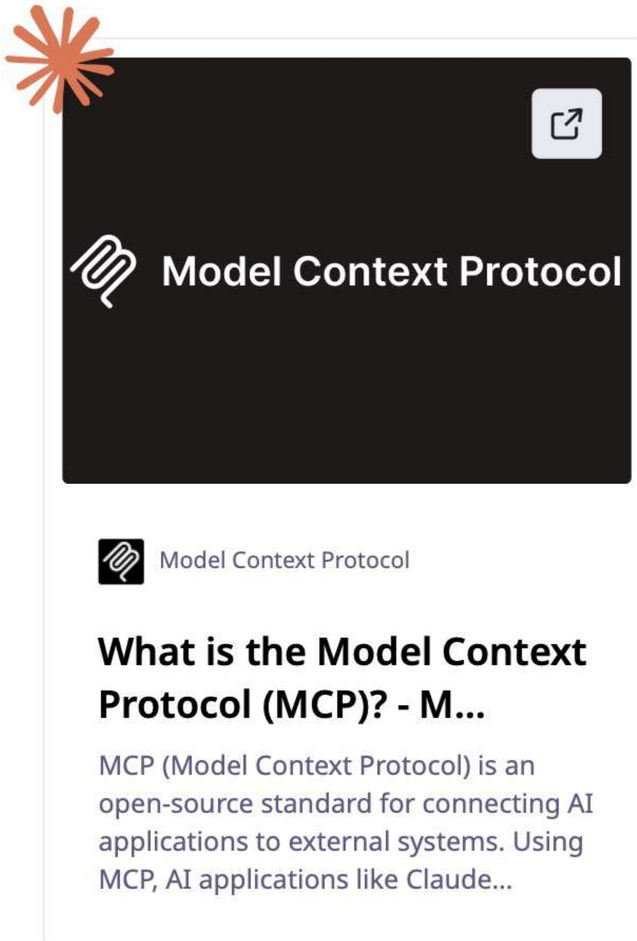
- Curious
- Actively seek improvement
- Love sharing

Why this workshop ?

- To have fun
- To learn more about MCP & AI
- To share my learnings
- To learn from the attendees
- To make it easier to take action

MCP

- Model Context Protocol
- Open-source standard for connecting AI applications to external system
- Shared by anthropic in November 2024
- Pretty well documented

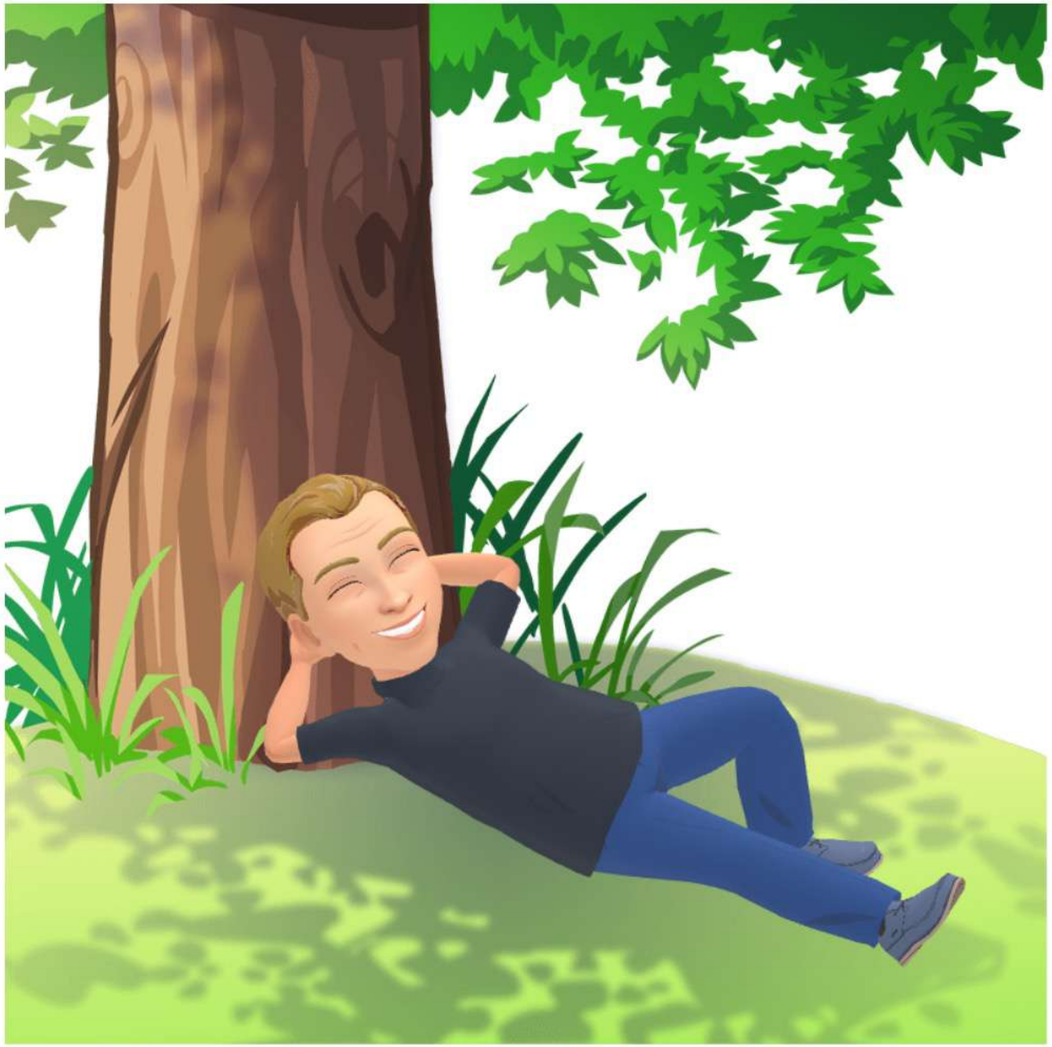


What for ?

- An API ...
- ... for AI Agents
- Allow agents to:
  - Read files, database
  - Call apis
  - Execute actions

At the roots

- Transport Layer
- Data layer



Transport layer

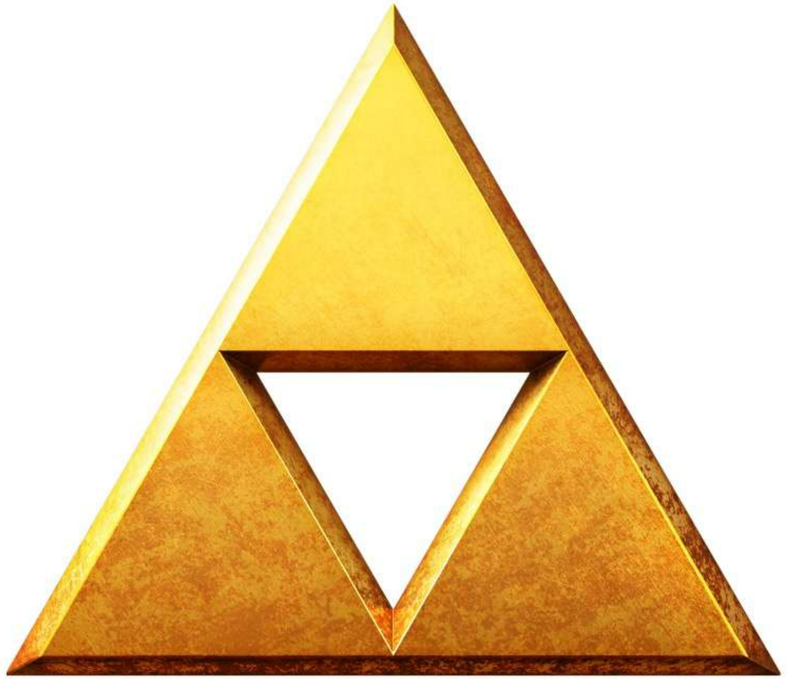
- StdIO
- Streamable HTTP

Data layer

- Json RPC 2-0
- JsonSchema

Server Primitives

- Resources
- Tools
- Prompts



Resources

- For reads
- Static or dynamic
- Discovery
- Text or binary
- Can subscribe to updates

Tools

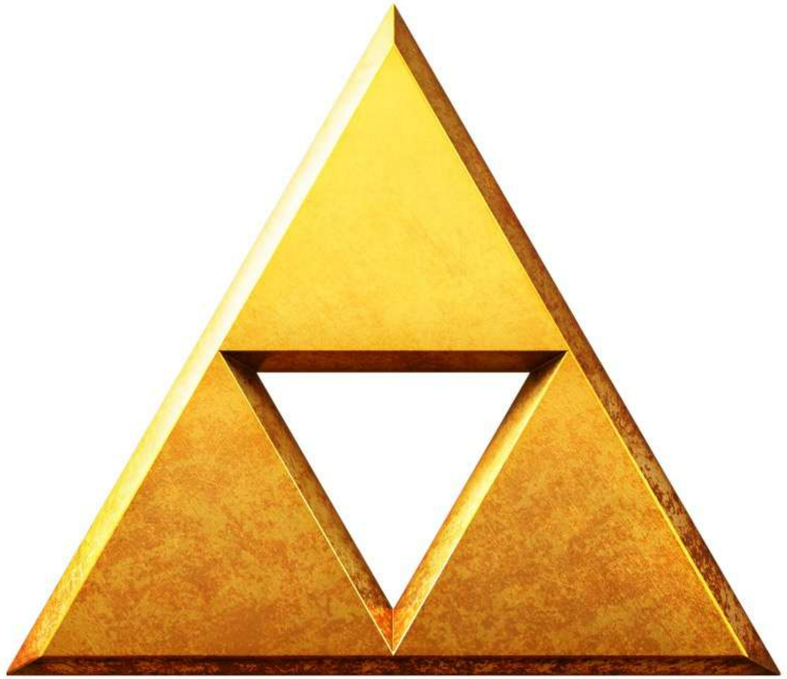
- Action executions
- Discovery
- Typed Schem
- Structured result
- Human approval advised

Prompts

- Reusable templates
- Discovery
- Dynamic arguments
- Usages: slash commands / workflows / standardized business instructions

Client Primitives

- Elicitation
- Roots
- Sampling



Elicitations

- On-demand interaction
- Dynamic forms
- Blocking flow
- User control
- *Confirmation, missing parameter, etc.*

Roots

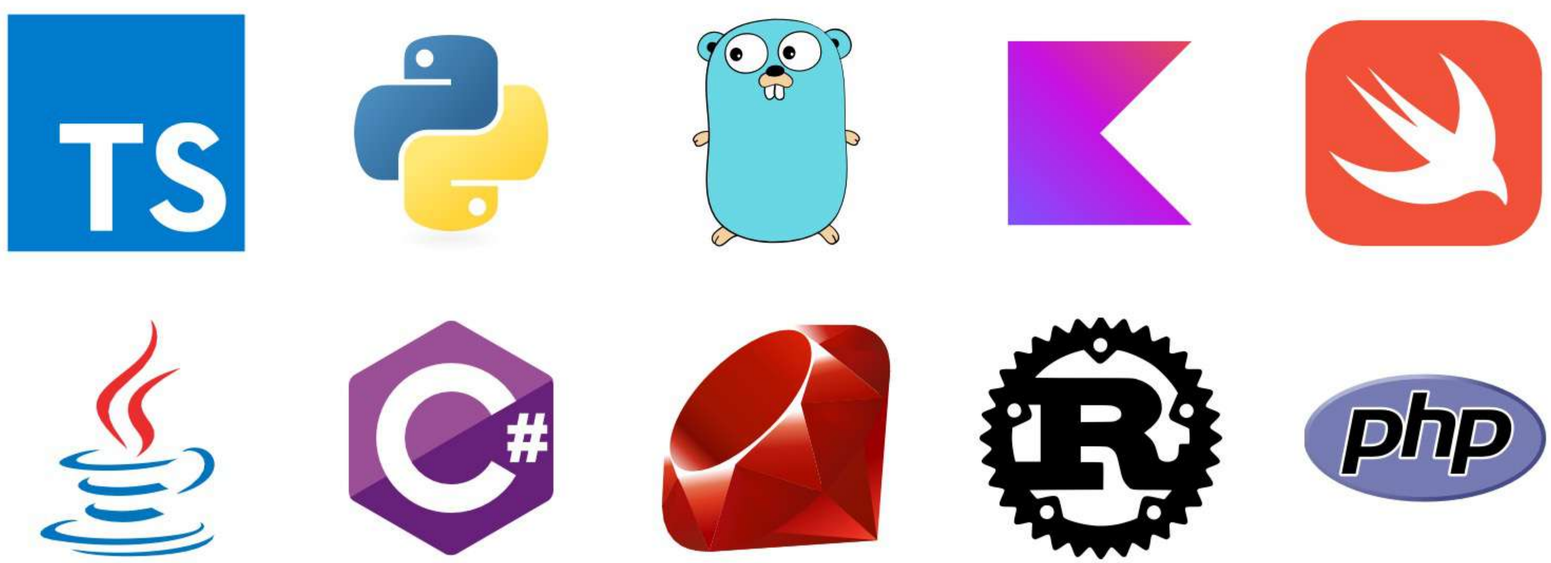
- Context URI
- Explicit boundaries
- Discovery
- Not security
- *Code workspace, project folder*

Sampling

- Ask back to the agent
- Structured query
- Human keep control
- Reusable response
- *Resume a file, classify data, etc.*

Building blocks

- MCP Inspector, to debug
- Implement it from scratch
- or use a SDK available for

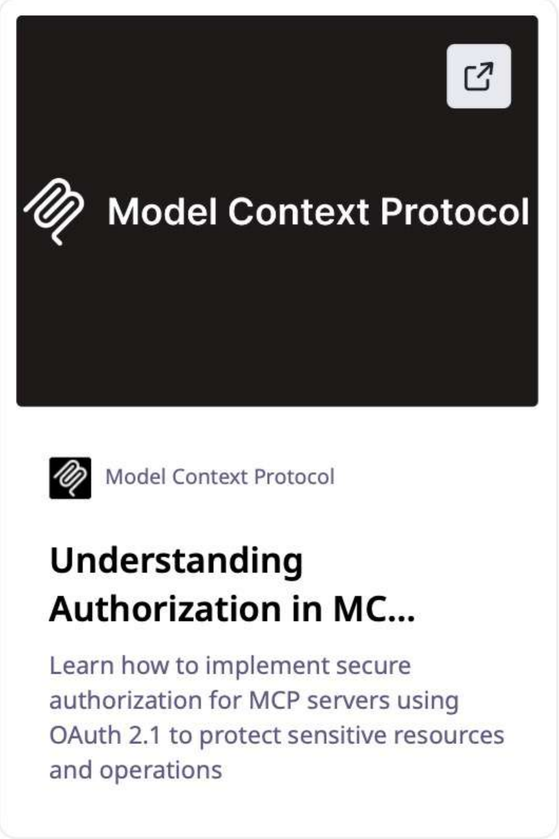


Caveat

- Partial support depending on the calling agent
- No security out of the box
- Protocol subject to evolution
- No execution guarantee (the agent make the decision)

Let's get started !

1. Install the inspector
2. Create and connect to a server
3. Request data from a resource
4. Use a tool to write to console
5. Create a wonderful prompt

To go further

- Build a MCP Client
- Remote MCP Server & authentication